

Tumar — Hackathon submission packet

A non-custodial family investment vault on Solana. Multi-currency on-ramp (KZTE, USDT, USDC, Palm USD), real-world assets out (Backed Finance xStocks, gold, jitoSOL), private deposits via Cloak, and a fully-offline Kazakh / Russian portfolio assistant on macOS via Tether QVAC. Mainnet program live: `HfCmnXggSF2tVQkCrEdPNjUTBYvvC8tgbebXES2sp24Y` . Live URL: web-phi-drab-54.vercel.app · alias `tumar.app` .

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One product, six side tracks

TRACK	PRIZE	DECK	WHY WE FIT
Cloak 5,010 USDC · winners May 14	\$5,010	tumar-deck-cloak.pdf	Privacy is load-bearing for cross-border family money. Three Cloak surfaces in production: private buy via <code>swapWithChange</code> , shielded deposit, family-level audit via <code>deriveDiversifiedViewingKey</code> . Electron app decrypts shielded history offline.
Tether QVAC 10,000 USDT · winners May 13	\$10,000	tumar-deck-qvac.pdf	macOS app runs Qwen3 1.7B + Whisper Tiny entirely on-device via QVAC. Russian-speaking grandmother queries family vault in Kazakh, no internet, no API key, no data leaves the laptop.
Palm USD 10,000 PUSD · winners May 27	10,000 PUSD	tumar-deck-palm.pdf	PUSD as a first-class deposit asset for the UAE-side Muslim diaspora. Non-freezable + halal + zero-yield is exactly what a multi-generation family vault needs. Direct-transfer path detects PUSD when no Jupiter route exists.
Superteam KZ general 4,000 USDG · winners May 29 · KZ-only	4,000 USDG	tumar-deck-kz.pdf	Built in Almaty, for the Kazakh diaspora. Founder ambition: this is the product I want to keep building post-Frontier. Solana load-bearing for sub-cent remittance fees and 24/7 settlement vs. Freedom Broker's KZ-residency T+2.

S1lkPay x KZ 3,000 USDG · winners May 29 · KZ-only	3,000 USDG	tumar-deck-s1lkpay.pdf	Cross-border tenge remittance, programmable. Tumar handles on-chain savings, S1lkPay handles spend. KZTE as first-class deposit; the natural off-ramp is exactly S1lkPay's virtual card.
Metaforra x KZ 3,000 USDG · winners May 29 · KZ-only	3,000 USDG	tumar-deck-metaforra.pdf	RWA tokenization in production today. Vault holds Backed Finance Token-2022 xStocks (AAPLx, NVDAx, SPYx) + GLDx + jitoSOL in PDA-owned ATAs. Real estate is the natural next allocation slot.

Combined max prize pool: ~\$35,000 across six tracks. Each track is its own application form on superteam.fun/earn — submit separately.

What's in the box

FILE	WHERE	WHAT IT IS
tumar-founder-en-subs.mp4	~/Downloads/	3-minute founder intro by Aigerim. Russian audio, English subtitles burned in. Audio denoised + loudness-normalized + downscaled to 1080p.
7 x deck PDFs	~/tumar/decks/tumar-deck-*.pdf	One per side track + universal -all . Palm/Cloak/QVAC variants are 12 slides (within Palm USD's hard cap); others are 13.
This document	~/tumar/decks/tumar-submission.pdf	You're reading it. Use it as the master reference when filling out the six application forms.
Code repo	~/tumar/	Public repo. README and SUBMISSION.md cover technical depth for Cloak + QVAC.

Submission checklist

Today is **April 28–29, 2026**. Frontier deadline: **May 11, 23:59 UTC**. Submit to all six side tracks separately on superteam.fun/earn after filing the master Frontier entry on arena.colosseum.org.

PRE-SUBMISSION — DO ONCE, APPLIES TO ALL SIX

- ☐ **Master Frontier entry** filed on arena.colosseum.org. Side tracks REQUIRE this as a precondition — without the Colosseum entry URL, the side-track forms can't validate.
- ☐ **Repo public** at github.com/<you>/tumar. Cloak + Tether explicitly require public GitHub. Palm USD allows private but you'd email access to hello@palmusd.com.
- ☒ **Mainnet program deployed**: `HfCmnXggSF2tVQkCrEdPNjUTBYvvC8tgbexXES2sp24Y`.
- ☒ **Live URL**: `web-phi-drab-54.vercel.app`.
- ☒ **macOS .dmg**: `pnpm --filter family build:dmg` → `apps/family/dist/Tumar-Family-0.1.0-arm64.dmg`.
- ☒ **Founder video**: `~/Downloads/tumar-founder-en-sub.s.mp4`.
- ☐ **Demo video** (≤5 min, screen recording with voiceover). The narration script is in this document — final section.

PER-TRACK CHECKLIST

TRACK	LISTING SLUG ON SUPETEAM.FUN/EARN	WHAT TO ATTACH
Cloak	<code>cloak-track</code>	Demo video URL · public repo URL · Colosseum entry URL · live URL · <code>tumar-deck-cloak.pdf</code>
Tether QVAC	<code>tether-frontier-hackathon-track</code>	Demo video URL · public repo URL · Colosseum entry URL · live URL or .dmg link · <code>tumar-deck-qvac.pdf</code>
Palm USD	<code>palm-usd-x-superteam-uae-solana-builders-1</code>	Demo · repo (or invite hello@palmusd.com) · Colosseum URL · live URL · <code>tumar-deck-palm.pdf</code>
Superteam KZ general	<code>side-track-superteam-kazakhstan</code>	Colosseum URL · demo · GitHub · KZ affiliation note · <code>tumar-deck-kz.pdf</code>
S1lkPay × KZ	<code>superteam-kz-x-s1lkpay-frontier-side-track</code>	Colosseum URL · demo · GitHub · KZ affiliation · <code>tumar-deck-s1lkpay.pdf</code>
Metaforra × KZ	<code>superteam-kz-and-metaforra-frontier-side-track</code>	Colosseum URL · demo · GitHub · KZ affiliation · <code>tumar-deck-metaforra.pdf</code>

RISKS TO FLAG NOW

- Six application forms × ~30–45 min each ≈ 4 hours of pure form-filling. Plan one block; don't do them last-minute.
- Helius free tier: 1M credits/day. One viral tweet during the demo period kills the proxy. Either upgrade Helius or add a fallback upstream before publishing the live URL widely.
- The Helius API key was pasted into chats. If you ever screen-record DevTools, scrub the Network tab. `vercel env rm` + regenerate is one minute.
- PUSD has no Jupiter route. The deposit code already handles this with the direct-transfer fallback, but a Palm USD judge testing with USDC will hit "TOKEN_NOT_TRADABLE". The pre-flight UI message ("you don't hold PUSD yet — get some at palmusd.com first") is already in place, but worth flagging in the demo.

Per-track application blurbs

~80–110 words each. Copy the relevant one into the application form's "Why does your project fit this track?" / "Project description" field.

CLOAK TRACK

Tumar is a family investment vault for the Kazakh diaspora — workers in the UAE, Korea, Turkey, Germany pooling money home into one Solana vault that holds real assets (xStocks, gold, jitoSOL) instead of evaporating tenge. Privacy is load-bearing: a public ledger of "who funds whose family" is unacceptable. Tumar uses Cloak across three surfaces — private buy (`swapWithChange` via Orca), shielded deposit (relay-submitted, on-chain link broken at the pool boundary), and family-level audit via vault-scoped `deriveDiversifiedViewingKey`. The Electron desktop app decrypts shielded history offline. Live on mainnet — every Groth16 proof in the demo is a real proof, not a mock.

TETHER QVAC TRACK

Tumar's family-finance app runs entirely on-device through Tether QVAC. The user is a Russian / Kazakh-speaking grandmother in a Kazakhstan village with patchy internet who doesn't trust cloud apps with her family's money. The macOS Electron app loads Qwen3 1.7B via `@qvac/llm-llamacpp` and Whisper Tiny via `@qvac/transcription-whispercpp`, both Metal-accelerated, both offline. She presses-and-holds the mic, speaks in Kazakh — "*Aidana bul ay qansha saldy?*" — and the local model answers in Kazakh, grounded in the on-chain contribution data for her family's vault. After model load: zero network calls. We unplug ethernet during the demo.

PALM USD TRACK

Tumar treats Palm USD as a first-class deposit asset for the UAE side of the Kazakh family corridor — the side where Muslim families won't accept USDC's interest exposure. PUSD's three properties (non-freezable SPL, fully-reserved Sharia-compliant basket, zero yield by construction) are exactly what a multi-generation family vault needs: nobody can freeze the app's stake, the dollar that lands is the dollar that's there a year later, and the structure is halal by default. PUSD has no Jupiter route at the time of writing, so we built a separate direct- `transferChecked` path that detects PUSD and routes correctly without forcing a swap. Live on mainnet.

SUPERTEAM KZ — GENERAL REGIONAL TRACK

Tumar is a family investment vault for the Kazakh diaspora — built in Almaty, for the workers in Korea sending tenge home that loses 11% a year, for the families saving for an Almaty apartment against 22% mortgage rates, for the Astana grandmother who wants to see what her son in Dubai actually contributed this month. One Solana vault, multi-currency on-ramp (KZTE / USDT / USDC / PUSD), real assets out (Apple, Nvidia, S&P 500, gold, staked SOL via Backed Finance xStocks). Solana is load-bearing: sub-cent fees make \$300 monthly remittances economical, Token-2022 makes xStocks possible, and ALTs make our compose-everything flows fit. Mainnet program live. Real founder ambition; this is the product I want to keep building after Frontier.

SUPERTEAM KZ x S1LKPAY

Cross-border tenge remittance, programmable. Tumar lets a worker abroad send into a single family vault on Solana — KZTE deposits straight from inside Kazakhstan, USDT from Asia and the Gulf, USDC from the US, all flowing through the same per-asset deposit path with sub-second settlement and sub-cent fees. The vault holds real assets (xStocks, gold, jitoSOL) and tenge stable as a withdrawal-side reserve. The natural off-ramp is exactly S1lkPay's strength: a virtual card backed by KZTE balance, spendable at any merchant. Tumar handles the on-chain savings layer, S1lkPay handles the spend — together that's the full cross-border family money loop.

SUPERTEAM KZ x METAFORRA

Real-world asset tokenization built into the family-money product itself. Tumar's vault doesn't just sit on stablecoins — it holds actual tokenized US equities (AAPLx, NVDAx, SPYx via Backed Finance Token-2022 xStocks with full transfer-hook compliance), tokenized gold (GLDx), and staked SOL (jitoSOL via Jito). Every position is in the vault's own PDA-owned ATA, readable by anyone, redeemable at the issuer. The architecture extends cleanly to fractionalized real estate (Astana / Almaty property tokens) as the natural next slot — same allocation system, same per-asset deposit path. Web2-simple onboarding via QR plus offline desktop app; Web3 capabilities underneath. Mainnet live.

Demo video — narration script

~4:30 talking-head + screen recording. Aigerim narrates over the actual app. Don't cut between every sentence — let the screen breathe. The founder intro video is already filmed and lives at `~/Downloads/tumar-founder-en-subs.mp4` ; this script is for the separate product demo video judges will watch alongside it.

[0:00 — talking head, 10s]

Hi, I'm Aigerim. I built Tumar — a family investment vault on Solana for the Kazakh diaspora. Let me show you what it does, and why I made the decisions I made.

[0:10 — over a vault page, 30s]

The tenge has been losing value against the dollar for years. Mortgage rates here are around twenty-two percent. Anyone with a US brokerage account has been compounding the whole time, and from Kazakhstan, in tenge, there's no clean way to get in. Families like mine are split between Kazakhstan, Korea, the Gulf, Germany — and there's no shared place to actually pool money and own real assets together.

[0:40 — switch to vault page, 30s]

This is a real family vault on Solana mainnet. The vault is its own on-chain account, owned by a program PDA. Right now it holds real NVDAx — Backed Finance's tokenized Nvidia stock — sitting in the vault's own associated token account, plus a little leftover USDC. Everything you see here is on-chain truth: we read the vault's actual token accounts, multiply by spot prices from Jupiter, and show the real dollar value. No fake allocation math.

[1:10 — Deposit → Per asset, 50s]

Here's how a family member deposits as the asset they want the vault to hold. Pick NVDAx, enter the USDC amount, sign two transactions. Jupiter swaps the USDC into NVDAx, then we transfer the asset into the vault's ATA and record the contribution on-chain. I split this into two transactions on purpose — Jupiter's swap, plus Token-2022 transfer hooks for xStocks, plus our ATA creation, plus the contribution record, do not all fit under Solana's 1232-byte packet limit in one v0 transaction. Two simple transactions are more reliable than one heroic one. The vault is stablecoin-agnostic — USDC, USDT, the Kazakh tenge stablecoin, and Palm USD all flow through the same per-asset path.

[2:00 — /buy, run a private buy, 50s]

Privacy was non-negotiable. Sending money home shouldn't be a permanent public record. This is the private buy flow, built on the Cloak SDK. Your USDC shields into a UTXO pool — Groth16 proof generated client-side in seconds — and Cloak's relay submits the swap on your behalf. The asset lands in your wallet with no on-chain link to where the USDC came from. Your employer doesn't see it. Anyone reading your wallet on Solscan doesn't see it. Only the people you actually told. I integrated swap-with-change through Orca and handled the stale-state edge case — when the merkle tree gets out of sync with the relay, the user gets a clear "reset state" button next to the error instead of an opaque hang.

[2:50 — desktop app, 60s]

This is the part I'm proudest of. A lot of older relatives in Kazakhstan don't have reliable

internet, and they don't trust cloud apps with their family's money. So I built a macOS app that runs the full AI stack on-device, through Tether's QVAC SDK. The language model — Qwen3 1.7B — runs locally via QVAC's Llamacpp engine. Whisper handles voice transcription, also through QVAC, also on-device. I can ask in Kazakh — "Aidana bul ay qansha saldy?" — how much did Aidana put in this month — and the app transcribes the audio, the local LLM reads the on-chain contribution data for the family's vault, and answers back in Kazakh. No internet. No API key. No subscription. No data leaving the laptop. The grandmother in the village gets to actually see her family's books, in her own language, with zero surveillance.

[3:50 — back to web, 25s]

Multiple stablecoins on the deposit side, by design. The Kazakh tenge stablecoin so the family at home can deposit in their own currency. USDT for the diaspora — that's what construction workers in Asia and the Gulf actually hold. USDC for everyone in the US flow. And Palm USD as a first-class option for Muslim families who care about Sharia-compliance — fully reserved, non-freezable, no interest. PUSD has no Jupiter route yet, so I built a separate direct-transfer path that detects PUSD and routes it correctly without trying to swap.

[4:15 — talking head close, 15s]

The Anchor program is live on Solana mainnet. The web app is at tumar.app. The desktop app is open source. Tumar means amulet in Kazakh — the small charm a grandmother sews into a baby's clothes before they go out into the world. That felt like the right name. Thanks for watching.